Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Fixed mass of an ideal gas contained in a 24.63 L sealed rigid vessel at 1 atm is heated from $-73^{\circ}\mathrm{C}$ to $27^{\circ}\mathrm{C}$. Calculate change in gibb's energy if entropy of gas is a function of temperature as $\mathrm{S}=2+10^{-2}\ \mathrm{T}(\ \mathrm{J/K}):(\text{ Use } 1\,\mathrm{atm\,L}=0.1\ \mathrm{kJ})$

(A) $1231.5\ \mathrm{J}$

(B) $1281.5\ \mathrm{J}$

(C) $781.5\ \mathrm{J}$

(D) 0

Q.2

For the reaction taking place at certain temperature $\mathrm{NH_2COONH_4}(\ \mathrm{s}) \rightleftharpoons 2\mathrm{NH_3}(\ \mathrm{g})+\mathrm{CO_2}(\ \mathrm{g})$ if equilibrium pressure is $3\mathrm{X}$ bar then $\Delta_{\mathrm{r}}\mathrm{G}^{\circ}$ would be

(A) $-\mathrm{RT}\ln 9-3\mathrm{RT}\ln \mathrm{X}$

(B) $\mathrm{RT}\ln 4-3\mathrm{RT}\ln \mathrm{X}$

(C) $-3\mathrm{RT}\ln \mathrm{X}$

(D) None of these

Q.3

Standard Gibb's free energy change for isomerization reaction cis-2-pentene $\rightleftharpoons \mathrm{trans}-2$ -pentene is $-3.67\ \mathrm{kJ/mol}$ at $400\ \mathrm{K}$. If more trans-2pentene is added to the reaction vessel, then

(A) more cis-2-pentene is formed

(B) equilibrium remains unaffected

(C) additional trans-2-pentene is formed

(D) equilibrium is shifted in forward direction

Q.4

The dissolution of KCl in water is endothermic yet it dissolves in water spontaneously. Which one of the following best explains this behaviour?

(A) Endothermic processes are energetically favoured.

(B) The electrostatic bonds between the ions are not too strong.

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(C) Energy changes have nothing to do with the dissolution processes.

(D) The entropy driving force causes the dissolution.

Q.5

What is the free energy change for the conversion of 1 mole of water into steam at $373.2\ \mathrm{K}$. The heat of vaporization $(\Delta \mathrm{H_v})$ of water of $373.2\ \mathrm{K}$ is $9.1 \mathrm{kcal mol^{-1}}$. The entropy change is $25.5 \mathrm{cal/mol deg}$.

(A) $-401.6 \mathrm{cal/mol}$

(B) $-416.6 \mathrm{cal/mol}$

(C) $516.5 \mathrm{cal/mol}$

(D) $-516.5 \mathrm{cal/mol}$

Q.6

A certain reaction is non spontaneous at $298\ \mathrm{K}$. The entropy change during the reaction is 121 $\mathrm{JK^{-1}}$. Is the reaction is endothermic or exothermic ? The minimum value of $\Delta \mathrm{H}$ for the reaction is

(A) endothermic, $\Delta \mathrm{H} = 36.06\ \mathrm{kJ}$

(B) exothermic, $\Delta \mathrm{H} = -36.06\ \mathrm{kJ}$

(C) endothermic, $\Delta \mathrm{H} = 60.12\ \mathrm{kJ}$

(D) exothermic, $\Delta \mathrm{H} = -60.12\ \mathrm{kJ}$

Q.7

The favourable conditions for a spontaneous reaction are

(A) $\mathrm{T\Delta S} > \Delta \mathrm{H}, \Delta \mathrm{H} = +\mathrm{ve}, \Delta \mathrm{S} = +\mathrm{ve}$

(B) $\mathrm{T\Delta S} > \Delta \mathrm{H}, \Delta \mathrm{H} = +\mathrm{ve}, \quad \Delta \mathrm{S} = -\mathrm{ve}$

(C) $\mathrm{T\Delta S} = \Delta \mathrm{H}, \Delta \mathrm{H} = -\mathrm{ve}, \quad \Delta \mathrm{S} = -\mathrm{ve}$

(D) $\mathrm{T\Delta S} = \Delta \mathrm{H}, \Delta \mathrm{H} = +\mathrm{ve}, \quad \Delta \mathrm{S} = +\mathrm{ve}$

Q.8

The standard Gibb's free energy change, $\Delta \mathrm{G^\circ}$ is related to equilibrium constant, $\mathrm{K_P}$ as

(A) $\mathrm{K_P} = -\mathrm{RT} \ln \Delta \mathrm{G^\circ}$

(B) $\mathrm{K_P} = \left[\frac{\mathrm{e}}{\mathrm{RT}}\right]^{\Delta \mathrm{G^\circ}}$

(C) $\mathrm{K_P} = -\frac{\Delta \mathrm{G}}{\mathrm{RT}}$

(D) $\mathrm{K_P} = \mathrm{e}^{-\Delta \mathrm{G^\circ}/\mathrm{RT}}$

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Q.9

The standard free energies of formation of $\mathrm{H_2S(g)}$ and $\mathrm{CdS(s)}$ at $1000^{\circ}\mathrm{C}$ are $-49.0\mathrm{kj/mol}$ and $-127.2\ \mathrm{kJ/mol}$, respectively. Use these data to predict whether $\mathrm{H_2(g)}$ will reduce CdS(s) to metallic Cd at this temperature

(A) $\Delta \mathrm{G} = -78.2\ \mathrm{kJ/mol}$ and $\mathrm{H_2}$ reduces CdS

(B) $\Delta \mathrm{G} = -39.1\ \mathrm{kJ/mol}$ and $\mathrm{H_2}$ reduces CdS

(C) $\Delta \mathrm{G} = 0\ \mathrm{kJ/mol}$ and the reaction is at equilibrium

(D) $\Delta \mathrm{G} = +78.2\ \mathrm{kJ/mol}$ and the reaction is not feasible

Q.10

For the equilibrium $\mathrm{H_2O(l) \rightleftharpoons H_2O(g)}$ at $1\ \mathrm{atm}$ and $298\ \mathrm{K}$

(A) Standard free energy change is equal to zero $(\Delta^{\circ}\mathrm{G} = 0)$

(B) Free energy change is less than zero $(\Delta G < 0)$

(C) Standard free energy change is less than zero $(\Delta G^{\circ} < 0)$

(D) Standard free energy change is greater than zero $(\Delta G^{\circ} > 0)$

Q.11

A diatomic ideal gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from V to $32V$, the efficiency of the engine?

(A) 0.5

(B) 0.75

(C) 0.99

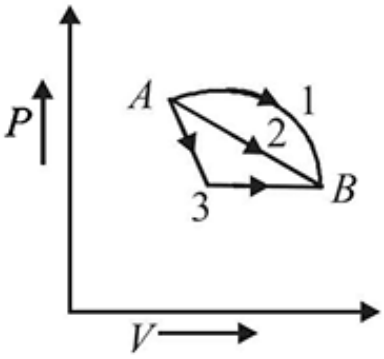
(D) 0.25

Q.12

An ideal gas goes from state A to state B via three different processes as indicated in the P-V diagram:
If $\mathrm{Q_1, Q_2, Q_3}$ indicate the heat a absorbed by the gas along the three processes and $\mathrm{\Delta U_1, \Delta U_2, \Delta U_3}$

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indicate the change in internal energy along the three processes respectively, then



(A) $Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$

(B) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$

(C) $Q_1 = Q_2 = Q_3$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

(D) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

Q.13

In a thermodynamic process, fixed mass of a gas is changed in such a manner that the gas release $20\ \mathrm{J}$ of heat and $8\ \mathrm{J}$ of work was done on the gas. If the initial internal energy of the gas was $30\ \mathrm{J}$, the final internal energy will be

(A) 2 joule

(B) 18 joule

(C) 42 joule

(D) 58 joule

Q.14

The (S°) of the following substances are:
CH_4 (g) $186.2 JK^{-1}\ mol^{-1}$
O_2 (g) $205.2 JK^{-1}\ mol^{-1}$
CO_2
(g) $213.6 JK^{-1}\ mol^{-1}$
$H_2O(g) 69.9 . JK^{-1}\ mol^{-1}$
The entropy change (ΔS°) for the reaction $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ is:

(A) $-312.5\ \mathrm{J\ K^{-1}\ mol^{-1}}$

(B) $-242.8\ \mathrm{J\ K^{-1}\ mol^{-1}}$

(C) $-108.1\ \mathrm{J\ K^{-1}\ mol^{-1}}$

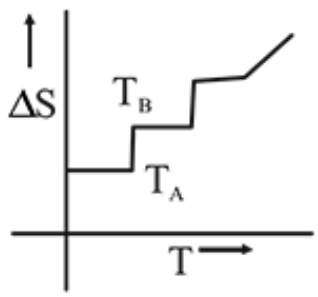
(D) $-37.6\ \mathrm{J\ K^{-1}\ mol^{-1}}$

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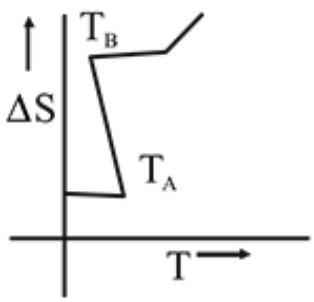
Q.15

If for a given substance melting point is T_B and freezing point is $\mathrm{T_A}$, then correct variation shown by, graph between entropy change and temperature is

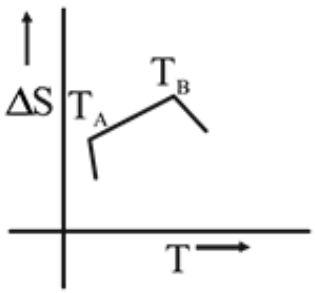
(A)



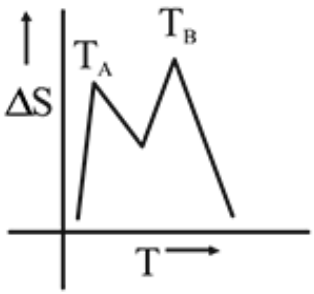
(B)



(C)



(D)



Q.16

The entropy of a sample of a certain substance increases by $0.836\ \mathrm{J\ K^{-1}}$ on adding reversibly $0.3344\ \mathrm{J}$ of heat at constant temperature. The temperature of the sample is:

(A) $2.5\ \mathrm{K}$

(B) $0.3\ \mathrm{K}$

(C) $0.016\ \mathrm{K}$

(D) $0.4\ \mathrm{K}$

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Q.17

One mole of an ideal gas is expanded isothermally and reversibly to half of its initial pressure. ΔS for the process in $\mathrm{JK^{-1}\ mol^{-1}}$ is $[\ln 2 = 0.693$ and $R = 8.314,\ \mathrm{J/(molK)}]$

(A) 6.76

(B) 5.76

(C) 10.76

(D) 8.03

Q.18

When $1\ \mathrm{kg}$ of ice at $0^\circ\mathrm{C}$ melts to water at $0^\circ\mathrm{C}$, the resulting change in its entropy, taking latent heat of ice to be $80\mathrm{cal}/^\circ\mathrm{C}$, is

(A) $273\mathrm{cal/K}$

(B) $253\mathrm{cal/K}$

(C) $263\mathrm{cal/K}$

(D) $293\mathrm{cal/K}$

Q.19

A monoatomic ideal gas goes through a process $p = p_0 - \alpha V$ where p_0 and α are positive constants and V is its volume. At what volume will the
entropy of gas be maximum?

(A) $\frac{5p_0}{6\alpha}$

(B) $\frac{p_0}{2\alpha}$

(C) $\frac{p_0}{4\alpha}$

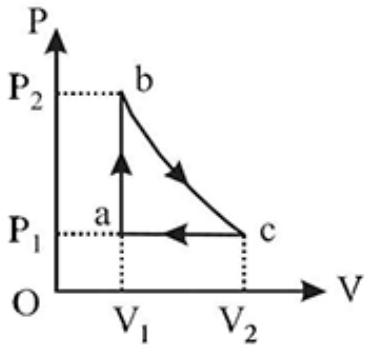
(D) $\frac{5p_0}{8\alpha}$

Q.20

Carbon monoxide is carried around a closed cycle abc in which bc isothermal process as shown in figure. The gas absorbs $7000\ \mathrm{J}$ of heat as its temperture increases from $300\mathrm{K}$ to $1000\ \mathrm{K}$ in going from a to b. quantity

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of heat rejected by the during the process ca is



(A) $4200\ \mathrm{J}$

(B) $5000\ \mathrm{J}$

(C) $9000\ \mathrm{J}$

(D) $9800\ \mathrm{J}$

Answers & Solutions

Q.1 Answer:

$781.5\ \mathrm{J}$

Solution:

At constant volume, $\frac{\mathrm{P_1}}{\mathrm{T_1}} = \frac{\mathrm{P_2}}{\mathrm{T_2}}$

$\Rightarrow \mathrm{P_2} = 1 \times \frac{300}{200} = \frac{3}{2}$

and $\mathrm{V_1} = 24.63\ \mathrm{L}$

for single phase

$\because \quad \mathrm{dG} = \mathrm{Vdp} - \mathrm{SdT}$

$\Delta \mathrm{G} = \mathrm{V}\Delta \mathrm{P} - \int \left(2 + 10^{-2}\ \mathrm{T}\right) \mathrm{dT}$

$= 1231.5 - 200 - \left(\frac{10^{-2} \times 50{,}000}{2}\right)$

$= 781.5\ \mathrm{J}$

Q.2 Answer:

None of these

Solution:

$\Delta \mathrm{G}^{\circ} = -\mathrm{RT} \ln \mathrm{K_p}; \mathrm{K_P} = (2\mathrm{x})^2 \mathrm{X} = 4\mathrm{x}^3$

$\Delta \mathrm{G}^{\circ} = -\mathrm{RT} \ln \left(4\mathrm{X}^3\right)$

$\Delta \mathrm{G}^{\circ} = -\mathrm{RT} \ln 4 - 3\mathrm{RT} \ln \mathrm{X}$

Q.3 Answer:

more cis-2-pentene is formed

Solution:

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If more trans-2-pentene is added, then 1 ts concentration in right hand side will increase. But in order to maintain the K constant, concentration of cis-2-pentene will also increase. Therefore more cis-2pentene will be formed.

Q.4 Answer:

The entropy driving force causes the dissolution.

Solution:

No solution available

Q.5 Answer:

$-416.6\mathrm{cal/mol}$

Solution:

$\Delta \mathrm{G} = \Delta \mathrm{H} - \mathrm{T}\Delta \mathrm{S}$
$\therefore \Delta G = 9100 - 373.2 \times 25.5 = -416.6\mathrm{cal}$
mol^{-1}

Q.6 Answer:

endothermic, $\Delta \mathrm{H} = 36.06\ \mathrm{kJ}$

Solution:

For non spontaneous reaction
$\Delta \mathrm{G} = +\mathrm{ve}$
$\Delta \mathrm{G} = \Delta \mathrm{H} - \mathrm{T}\Delta \mathrm{S}$ **and**
$\Delta \mathrm{S} = 121\mathrm{JK}^{-1}$
For $\Delta \mathrm{G} = +$ **ve**
$\Delta \mathrm{H}$ **has to be positive. Hence the reaction is endothermic.**

The minimum value of $\Delta \mathrm{H}$ **can be obtained**
by putting $\Delta \mathrm{G} = 0$
$\Delta \mathrm{H} = \mathrm{T}\Delta \mathrm{S} = 298 \times 121\ \mathrm{J}$
$= 36.06\ \mathrm{kJ}$

Q.7 Answer:

$\mathrm{T}\Delta \mathrm{S} > \Delta \mathrm{H}, \Delta \mathrm{H} = +\mathrm{ve}, \Delta \mathrm{S} = +\mathrm{ve}$

Solution:

$\Delta \mathrm{G} = \Delta \mathrm{H} - \mathrm{T}\Delta \mathrm{S}, \Delta \mathrm{H}$ **is** $+\mathrm{ve}$,
$\Delta \mathrm{S}$ **is** $+\mathrm{ve}; \mathrm{T}\Delta \mathrm{S} > \Delta \mathrm{H}$
for spontaneous process. It will make $\Delta \mathrm{G}, -\mathrm{ve}$

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Q.8 Answer:

$K_P = e^{-\Delta G^\circ/RT}$

Solution:

$\Delta G = -RT \ln K_p$ **or** $K_p = e^{\Delta G/RT}$

Q.9 Answer:

$\Delta G = +78.2\ kJ/mol$ and the reaction is not feasible

Solution:

$CdS + \mathbf{H}_2 \rightarrow Cd + H_2S$

$\Delta G = \Delta G(H_2S) - \Delta G(CdS)$

$= -49 - (127.2)$

$= +78.2\mathbf{kJ}$

Thus, (d)

Q.10 Answer:

Free energy change is less than zero $(\Delta G < 0)$

Solution:

For reaction $H_2O(l) \rightleftharpoons H_2O(g)$

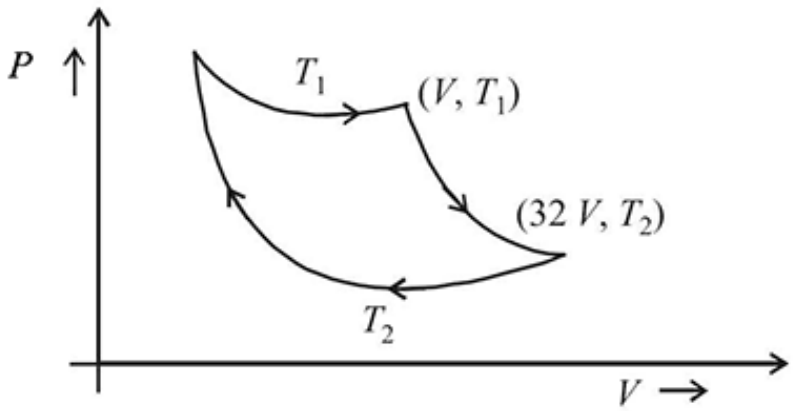
$\Delta n = 1$ **means positive**

so when Δn **is positive then** $\Delta G < 0$.

Q.11 Answer:

0.75

Solution:



We have, $TV^{g-1} =$ **constant** $\Rightarrow T_1V^{g-1} = T_2(32V)^{g-1}$

$\Rightarrow T_1 = (32)^{g-1} \cdot T_2$

For diatomic gas, $g = \frac{7}{5} \therefore g - 1 = \frac{2}{5}$

$\therefore T_1 = (32)^{\frac{2}{5}} \cdot T_2 \Rightarrow T_1 = 4T_2$

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Now, efficiency $= 1 - \frac{T_2}{T_1}$

$= 1 - \frac{T_2}{4T_2} = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$

Q.12 Answer:

$Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$

Solution:

Initial and final condition is same for all process $\Delta U_1 = \Delta U_2 = \Delta U_3$

from first law of thermodynamics $\Delta Q = \Delta U + \Delta W$

Work done $\Delta W_1 > \Delta W_2 > \Delta W_3$ **(Area of P.V. graph) So** $\Delta Q_1 > \Delta Q_2^2 > \Delta Q_3$

Q.13 Answer:

18 joule

Solution:

According to first law of thermodynamics, $\Delta Q = \Delta U + \Delta W$

$\Delta Q =$ **heat absorbed by gas** $\Delta W =$ **work done by gas.**

$-20 \text{ J} = \Delta U - 8 \text{ J}$

$\Delta U = -12 \text{ J} = U_{\text{Final}} - U_{\text{initial}}$

$U_{\text{initial}} = 30 \text{ J}$

$U_{\text{Final}} = 30 - 12 = 18 \text{ J}$

Q.14 Answer:

$-242.8 \text{ J K}^{-1} \text{ mol}^{-1}$

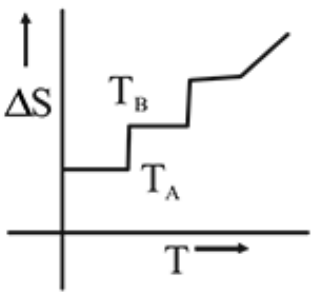
Solution:

$\Delta S^\circ = S^\circ CO_2 + 2 \times S^\circ H_2O - \left(S^\circ CH_4 + 2 \times S^\circ_{O_2}\right)$

$= (213.6 + 2 \times 69.9) - (186.2 + 2 \times 205.2)$

$= -242.8 \text{ J K}^{-1} \text{ mol}^{-1}.$

Q.15 Answer:



Solution:

For a pure substance T_A **and** T_B **represent the same temperature. Hence** A **is a correct choice.**

Q.16 Answer:

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$0.4\ \mathrm{K}$

Solution:

$\Delta S = \frac{\Delta H}{T}; T = \frac{\Delta H}{\Delta S} = \frac{0.3344}{0.836} = 0.4\ \mathrm{K}$

Q.17 Answer:

5.76

Solution:

For isothermal process $(\Delta T = 0)$

$$\Delta S = R \ln \frac{P_1}{P_2} = 8.314 \ln 2$$
$$= 8.314 \times 0.693 = 5.76$$

Q.18 Answer:

$293\mathrm{cal/K}$

Solution:

Change in entropy is given by $\mathrm{dS} = \frac{\mathrm{dQ}}{\mathrm{T}}$ **or** $\Delta \mathrm{S} = \frac{\Delta \mathrm{Q}}{\mathrm{T}} = \frac{\mathrm{mL_f}}{273}$

$\Delta \mathrm{S} = \frac{1000 \times 80}{273} = 293\mathrm{cal/K}$

Q.19 Answer:

$\frac{5p_0}{8\alpha}$

Solution:

$ds = nC_v dT + PdV = 0$

$nR\frac{dT}{dV} + (p_0 - \alpha V) = 0$

$pV = nRT$

$p_0 V - \alpha V^2 = nRT$

$p_0 - 2\alpha V = nR\frac{dT}{dV}$

$-(p_0 - \alpha V)(\gamma - 1) = p_0 - 2\alpha V$

$-p_0(\gamma - 1) + \alpha(\gamma - 1)V = p_0 - 2\alpha V$

$p_0 V = \alpha V(\gamma + 1)$

$V = \frac{p_0 \gamma}{\alpha(\gamma+1)}$

$V = \frac{p_0 \times \frac{5}{3}}{\alpha\left(\frac{5}{3}+1\right)} = \frac{5p_0}{8\alpha}$

Q.20 Answer:

$9800\ \mathrm{J}$

Solution:

For path ab : $(\Delta U)_{ab} = 7000\ \mathrm{J}$

By using $\Delta U = \mu C_V \Delta T$

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$7000 = \mu \times \frac{5}{2} R \times 700 \Rightarrow \mu = 0.48$

For path ca :

$(\Delta Q)_{ca} = (\Delta U)_{ca} + (\Delta W)_{ca}$

$\because (\Delta U)_{ab} + (\Delta U)_{bc} + (\Delta U)_{ca} = 0$

$\because 7000 + 0 + (\Delta U)_{ca} = 0 \Rightarrow (\Delta U)_{ca} = -7000J \ldots$ **(ii)**

Also $(\Delta W)_{ca} = P_1 (V_1 - V_2) = \mu R (T_1 - T_2)$

$= 0.48 \times 8.31 \times (300 - 1000) = -2792.16 \text{ J}$

On solving equations (i), (ii) and (iii)

$(\Delta Q)_{ca} = -7000 - 2792.16 = -9792.16J \approx -9800 \text{ J}$

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